

Cold-atom comagnetometry via optical control of spin states

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Atomic-spin-based comagnetometers are powerful tools for precision sensing and tests of fundamental physics. Compared with the widely used gas-cell comagnetometer systems, cold-atom systems offer access to much shorter distance scales and allow the implementation of optical quantum control techniques. However, to realize long spin coherence times with cold atoms, it is necessary to use diamagnetic atoms and overcome decoherence induced by light shifts. Here we demonstrate a cold-atom comagnetometer based on the nuclear spins of ^{171}Yb (spin-1/2) and ^{173}Yb (spin-5/2), jointly trapped in an optical lattice. Vector light shifts are suppressed by enforcing linear polarization of the lattice, whereas tensor shifts in ^{173}Yb are suppressed via the use of a Schrödinger cat state. This enables simultaneous Ramsey interferometry on both isotopes with a spin coherence time of 60 s. We achieve a magnetic noise suppression factor exceeding 3×10^4 , and determine the ^{171}Yb – ^{173}Yb ratio of nuclear magnetic moments to be $-0.726076(3)$ with a 4-ppm precision. Our results establish a new cold-atom platform for spin-based sensing and open pathways towards quantum-enhanced searches for physics beyond the Standard Model.

Atomic-spin-based sensors are indispensable in quantum information science¹ and precision measurement^{2,3}. Paramagnetic atoms, whose spins carry Bohr magnetons, underpin some of the most sensitive magnetometers⁴. By contrast, diamagnetic atoms possess spins originating solely from their nuclei. These nuclear spins, governed by much smaller nuclear magnetons, exhibit significantly longer coherence times⁵, making them ideal for sensing interactions beyond magnetism. Nuclear magnetic resonance is a prominent example of such sensing.

To suppress magnetic-field noise, a comagnetometer uses two atomic species with distinct gyromagnetic ratios, colocated and measured simultaneously⁶. The ratio of their Larmor frequencies is intrinsically insensitive to magnetic-field fluctuations and retains sensitivity to non-magnetic interactions. Such systems have demonstrated the exceptional ability to detect energy shifts at the picohertz level⁷.

Room-temperature gas-cell comagnetometers have been extensively developed, with combinations including the ^{85}Rb – ^{87}Rb paramagnetic pair⁸, paramagnetic–diamagnetic pairs K – ^3He (ref. 7) and Rb – ^{21}Ne (ref. 9), and

diamagnetic pairs ^3He – ^{129}Xe (refs. 9–11), Xe isotopes^{12,13} and Hg isotopes¹⁴. These platforms support a wide range of applications, from inertial sensing¹⁵ to precision tests of fundamental physics, including searches for electric dipole moments (EDMs)^{16–18}, monopole–dipole interactions^{12,13,19}, spin–gravity couplings^{7,8,20,21} and axion-like dark matter^{22,23}.

When probing for short-range interactions between gas atoms in a cell and a test object outside, the millimetre-thick cell wall imposes a limitation on sensitivity to interactions with ranges below $\sim 100\text{ }\mu\text{m}$. Moreover, because the atoms are distributed throughout the centimetre-sized cell, only a small fraction are near the wall and, thus, within the sensing range, which further dilutes the sensitivity to short-range interactions¹³. Cold-atom comagnetometers can overcome this limitation, as the test object can be placed inside the vacuum chamber in which the trap resides. The atom cloud confined in a laser trap can be reduced to scales of $1\text{--}100\text{ }\mu\text{m}$, enabling sensitivity to comparable interaction ranges. Furthermore, the absence of cell walls near the atoms and the test object eliminates related systematic effects.

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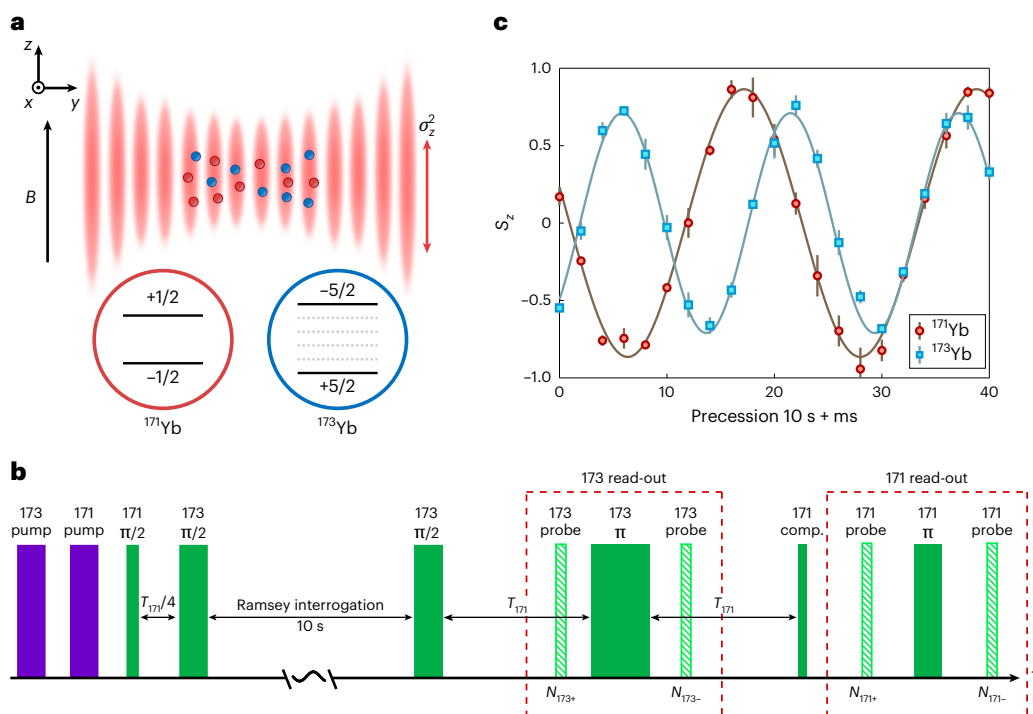


Fig. 1 | ^{171}Yb – ^{173}Yb cold-atom comagnetometer. **a**, Atoms of ^{171}Yb and ^{173}Yb are mixed and confined together in the optical lattice. A uniform field of 10–100 mG is applied to define the quantization axis in the vertical z direction. Although ^{171}Yb ($I = 1/2$) naturally has no tensor light shifts, ^{173}Yb ($I = 5/2$) is prepared into the cat state, which is immune to decoherence caused by the tensor shifts in the optical lattice. **b**, Measurement sequence of the simultaneous Ramsey interferometry. The arrangement of the 399-nm pump pulses (solid purple), 556-nm Rabi pulses

(solid dark green) and 556-nm probe pulses (hatched light green) are explained in detail in the main text. **c**, Ramsey signals at 10 s with a bias field at 50 mG, where red circles represent the precession signals of ^{171}Yb and blue squares correspond to those of ^{173}Yb . Error bars: one standard deviation ($1 - \sigma$) derived from three repeated measurements; solid curves: least-squares fittings with sinusoidal functions.

A prior implementation used two hyperfine states of paramagnetic ^{87}Rb with spin coherence times at several milliseconds²⁴. Signals of the two hyperfine states were combined to generate a single-species magnetometer based on nuclear spin with a coherence time of 1 s. More recently, a cold-atom system combining paramagnetic ^{87}Rb with diamagnetic ^{87}Sr was realized²⁵. Operating under magnetic fields of 100–1,000 G, this system enabled parts-per-million-level determination of the nuclear g -factor of ^{87}Sr , though its coherence time was ultimately limited to milliseconds by the ^{87}Rb component under strong magnetic fields.

Here we realize a comagnetometer based purely on nuclear spins consisting of ^{171}Yb (spin-1/2) and ^{173}Yb (spin-5/2), simultaneously confined in a one-dimensional optical lattice under a weak magnetic field (10–100 mG). A key challenge in such systems is the suppression of lattice-induced light shifts. We eliminate vector light shifts by enforcing linear polarization of the lattice beam. Tensor shifts are naturally absent in ^{171}Yb due to its spin-1/2 character. For the higher-spin ^{173}Yb , we apply quantum control techniques to prepare a Schrödinger cat state, whose precession frequency is rendered insensitive to tensor shifts. Using simultaneous Ramsey interferometry on both isotopes, we measure the Larmor-frequency ratio and demonstrate a spin coherence time of 60 s. We identify a novel systematic effect in the cat state stemming from the interplay between Zeeman and tensor light shifts (see the ‘Tensor shift’ section). By understanding and controlling both vector and tensor light shifts, we achieve a magnetic noise suppression factor exceeding 3×10^4 . We determine the ratio of nuclear magnetic moments between ^{171}Yb and ^{173}Yb with a precision of 4 ppm.

Results

Realization of a cold-atom comagnetometer

In the preparation of a cold-atom ensemble, ytterbium atoms containing both ^{171}Yb and ^{173}Yb (isotopic abundances of 14% and 16%,

respectively) exit an oven, pass through a Zeeman slower and are captured by a blue magneto-optical trap formed with a laser exciting the broad transition $6s^2^1S_0 \rightarrow 6s6p^1P_1$ at 399 nm. The atoms are subsequently cooled to a temperature of $\sim 20 \mu\text{K}$ with a green magneto-optical trap at 556 nm acting on the narrow transition $6s^2^1S_0 \rightarrow 6s6p^3P_1$, and then transferred to a movable (bus) optical dipole trap at 1,036.4 nm, which is a magic wavelength for the cooling transition²⁶. Carried by the bus optical dipole trap from the magneto-optical trap chamber to an adjacent science chamber across 0.35 m, the atoms arrive at their final destination: a horizontally directed, one-dimensional optical lattice also at 1,036.4 nm (Fig. 1a). With a beam power of 30 W and a waist of 25 μm , the lattice consists of trap sites with a depth of $\sim 2 \text{ mK}$. A total of about 2×10^4 atoms of ^{171}Yb and 2×10^4 atoms of ^{173}Yb are mixed together to form an ellipsoidal cloud with a diameter of 40 μm and a length of 80 μm , occupying approximately 160 lattice sites. The lifetime of the atoms in the optical lattice is observed to be 50 s.

The science chamber is surrounded by a set of coils and magnetic shields. A uniform magnetic field of 10–100 mG in the z direction is applied to define the spin quantization axis (Fig. 1a). In the 1S_0 , $F = 1/2$ ground state of ^{171}Yb , a transversely polarized state $|\psi_t\rangle = |F = 1/2, m_F = +1/2\rangle_g + e^{i\phi_1}|F = 1/2, m_F = -1/2\rangle_g$ is used to sense the precession phase ϕ_1 . For the ground state of ^{173}Yb ($I = 5/2$), it is necessary to prepare atoms into the cat state, $|\psi_c\rangle = |5/2, +5/2\rangle_g + e^{i\phi_2}|5/2, -5/2\rangle_g$ to achieve immunity to otherwise-significant decoherence caused by tensor light shifts in the optical lattice²⁷. We note that in the $\text{SU}(n)$ space, the cat state is structurally analogous to another non-classical quantum state, the NOON state²⁸. The cat state cannot simply be prepared with either optical pumping or a rotating magnetic field. Instead, it is generated using a 556-nm Rabi beam, a circularly polarized laser beam travelling in the x direction to drive the necessary Rabi transitions, with its approximately gigahertz

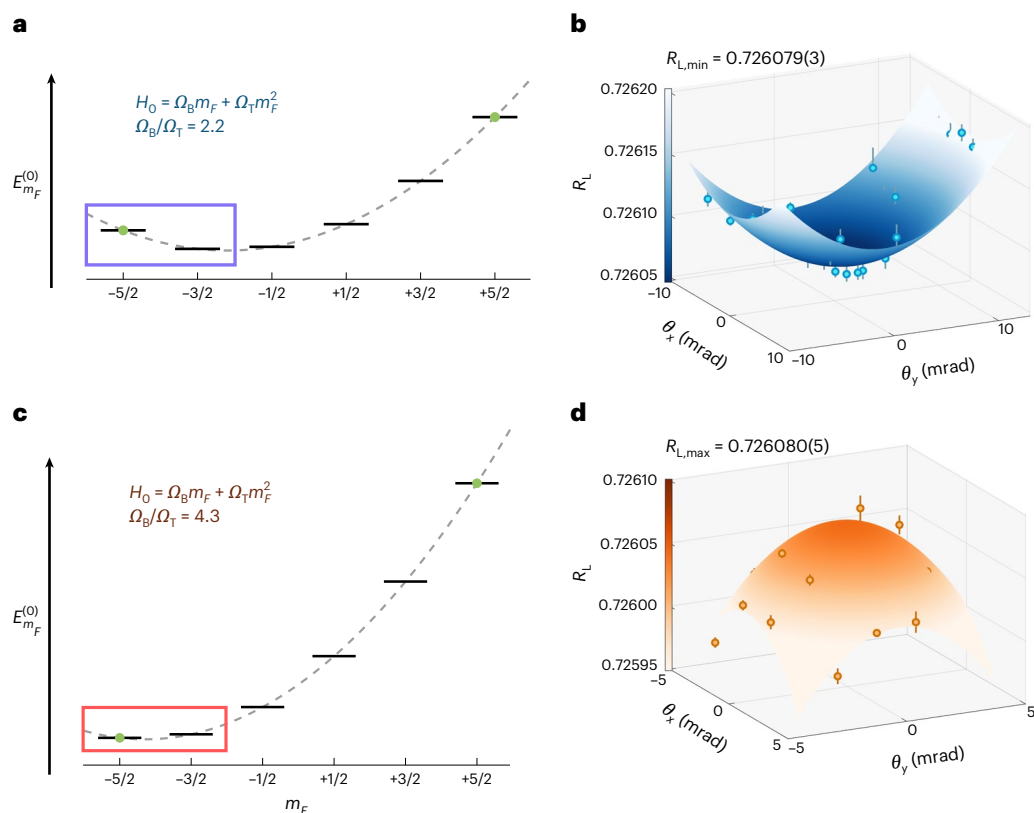


Fig. 2 | Combined effect of Zeeman and tensor shifts on the Larmor frequency (ω_{173}) of the cat state of ^{173}Yb and on $R_L = \omega_{171}/\omega_{173}$. ^{173}Yb atoms are populated in $|5/2, \pm 5/2\rangle_g$ (green dots). **a**, At $\Omega_B/\Omega_T = 2.2$, where $|5/2, -5/2\rangle_g$ is shifted higher than $|5/2, -3/2\rangle_g$, the σ_x term mixes the two states in the rectangle and pushes $|5/2, -5/2\rangle_g$ further up. This results in a lower frequency ω_{173} and a higher R_L value. **b**, Under 50 mG, corresponding to $\Omega_B/\Omega_T = 2.2$, R_L is mapped as the magnetic-field

orientation is scanned around the z direction. The higher R_L values in **a** result in a surface with positive curvatures. **c**, At $\Omega_B/\Omega_T = 4.3$, where $|5/2, -5/2\rangle_g$ is shifted below $|5/2, -3/2\rangle_g$, the σ_x term pushes $|5/2, -5/2\rangle_g$ further down. **d**, Under 100 mG, corresponding to $\Omega_B/\Omega_T = 4.3$. The curvature of the surface is found to reverse. Error bars: one standard deviation ($1 - \sigma$) derived from 50 repeated measurements; surfaces: least-squares fittings with quadratic surface functions.

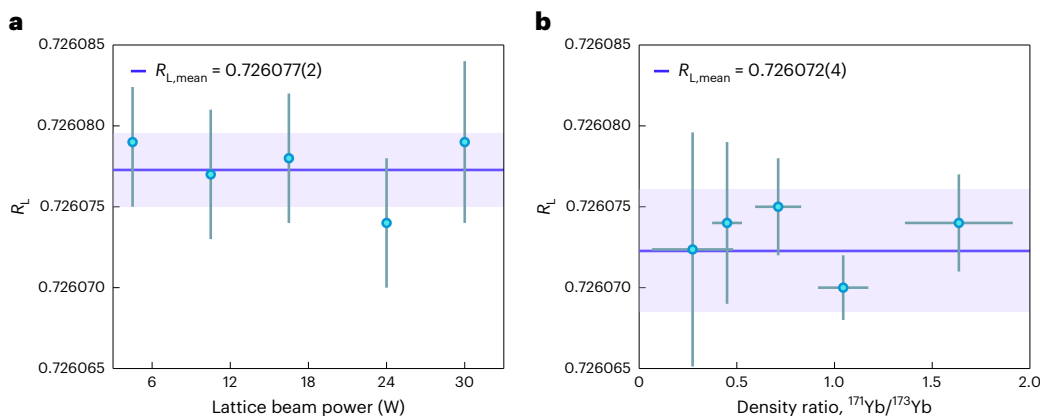


Fig. 3 | Investigating possible systematic effects. **a**, In a magnetic field of 100 mG, the lattice beam power is varied. **b**, With both magnetic field and lattice parameters fixed, the density ratio between ^{171}Yb and ^{173}Yb is varied. No

dependencies on these experimental parameters are observed. Error bars: one standard deviation ($1 - \sigma$) derived from 50 repeated measurements; solid lines: weighted averages.

detunings from the $6s^2\ ^1S_0 \rightarrow 6s6p\ ^3P_1$ transitions designed to induce a specific combination of vector and tensor light shifts²⁷. For convenience, the same Rabi beam is also used to drive Rabi transitions in ^{171}Yb , where it induces a pure vector shift in the spin-1/2 system. In this work, the induced Rabi transition frequency is $2\pi \times 0.4$ kHz for ^{173}Yb and $2\pi \times 3$ kHz for ^{171}Yb .

Ramsey interferometry is simultaneously implemented on both isotopes. Its measurement sequence is illustrated in Fig. 1b. Since all laser pulses are simultaneously shone on atoms of both isotopes, care

needs to be taken either to avoid crosstalks between the two isotopes or to compensate for any crosstalk effects. First, optical pumping at 399 nm along the z axis is applied to polarize ^{173}Yb and ^{171}Yb successively. Then, a $\pi/2$ pulse of the 556-nm Rabi beam is applied to drive the spin state of ^{171}Yb . The pulse also induces a small rotation ($-\pi/16$) on the spin of ^{173}Yb . After a quarter of the ^{171}Yb Larmor period ($T_{171}/4$), a $\pi/2$ pulse of the Rabi beam is applied to drive ^{173}Yb into the cat state, having little effect on ^{171}Yb because their spins are pointing along the Rabi beam direction at the time of the pulse. Afterwards, spins of both isotopes

Table 1 | Compilation of reported values for R_{NMM}

Measurement	R_{NMM}
Nuclear magnetic resonance on solid Yb (ref. 31)	−0.7278(16)
RF spectroscopy on Yb atoms in a beam ³²	−0.72610(4)
Comagnetometry on cold Yb atoms in a trap (this work)	−0.726076(3)

Values in the parentheses represent one standard uncertainty (1- σ).

are allowed to precess freely and accumulate phases over time. At the read-out stage, first on ^{173}Yb , the phase information is converted into population values by applying a $\pi/2$ pulse to the precessing cat state. The populations are measured as follows (see the ‘Atom sample detection’ section): the first probe pulse at 556 nm measures the population of $|5/2, +5/2\rangle_g$, yielding N_{173+} ; a subsequent π pulse exchanges the populations between $|5/2, +5/2\rangle_g$ and $|5/2, -5/2\rangle_g$; the second probe then captures the original population of $|5/2, -5/2\rangle_g$, denoted as N_{173-} . The differential signal is derived as $S_z = (N_+ - N_-)/(N_+ + N_-)$. Note that both the second $\pi/2$ pulse and the π pulse on ^{173}Yb unavoidably operate on ^{171}Yb , too. To compensate their effects on ^{171}Yb , an additional pulse is introduced so that the three pulses, each successive one delayed by a complete ^{171}Yb Larmor period (T_{171}), combine to rotate the spin of ^{171}Yb by $(12 + 1/2)\pi$. Finally, the populations of ^{171}Yb spin states are read out in a way similar to that of ^{173}Yb .

The resulting Ramsey fringes are displayed in Fig. 1c. The contrast of ^{173}Yb decays due to interspecies collision between ^{173}Yb and ^{171}Yb , reducing the coherence time of ^{173}Yb from thousands of seconds down to 60 s. Meanwhile, the collisions seem to have no degradation effects on ^{171}Yb .

By applying simultaneous Ramsey interferometry on both isotopes, we determine the precession phases, and derive the Larmor frequencies of ^{171}Yb (ω_{171}) as well as the cat state of ^{173}Yb (ω_{173}), both with a statistical sensitivity of $\sim 2\pi \times 2$ mHz in a single-shot measurement, primarily limited by spin-projection noise and residual light-shift fluctuations from the optical trap.

Under ideal conditions, the comagnetometer frequency ratio $R_L = \omega_{171}/\omega_{173}$ is insensitive to magnetic-field fluctuations. However, we find that light shifts induced by the high-power optical lattice could cause a significant perturbation to R_L .

Dependence on light shift

The light shift can be decomposed into scalar, vector and tensor terms, which are proportional to the zeroth, first and second powers of the magnetic quantum number m_F , respectively²⁹. The scalar shift causes identical displacements for all m_F states, thereby contributing no phase shifts in the evolution of angular momentum states. The vector shift, induced by circularly polarized light, acts as a Zeeman-like shift with its quantization axis along the wavevector of the light field. Given the linearly polarized optical lattice used in this work, the tensor shift aligned along the polarization direction emerges as the predominant effect.

Tensor shift. Tensor shifts are naturally absent in the ground state 1S_0 of ^{171}Yb ($I = 1/2$). For ^{173}Yb ($I = 5/2$), tensor shifts are also expected to be null in the cat state of 1S_0 , but instead are found in this work to affect the Larmor frequency in a significant and peculiar way. The tensor operator aligns along the linear polarization direction (z) of the lattice, which, due to imperfection, forms a small angle θ with the bias magnetic field B in the z direction. This misalignment causes a twist of the cat state, producing a frequency shift through a mechanism analogous to the alignment-to-orientation conversion effect in nonlinear Faraday rotation³⁰. Without loss of generality, assume the magnetic field has a small component in the transverse x direction, the Hamiltonian $\mathcal{H}$ arising from the combined magnetic and tensor fields can be expressed as

$$\mathcal{H} = \Omega_B(\sigma_z + \theta\sigma_x) + \Omega_T\sigma_z^2 \quad (\theta \ll 1). \quad (1)$$

Here σ_z and σ_x are the angular momentum matrices, with their terms corresponding to Zeeman effects. The σ_z^2 term describes the tensor effect. Ω_B and Ω_T are the corresponding coupling parameters.

Under constant optical trapping conditions, we scan the orientation of the bias magnetic field over two dimensions and map the Larmor-frequency ratio R_L . First, the magnetic field is set at 50 mG, corresponding to $\Omega_B/\Omega_T = 2.2$ (see the ‘Calibration of Ω_B/Ω_T ’ section). The measured R_L values fall onto a parabolic surface with a minimum extremum $R_{L,\text{min}} = 0.726079(3)$ (Fig. 2b). Next, at a higher magnetic field of 100 mG, where $\Omega_B/\Omega_T = 4.3$, the opening direction of the parabolic surface is found to be inverted, yielding a maximum $R_{L,\text{max}} = 0.726080(5)$ (Fig. 2d), with its value consistent with the minimum value obtained at 50 mG. The parabolic surface and its curvature inversion can be attributed to a second-order perturbation effect on the ^{173}Yb energy levels, with the transverse σ_x term treated as a perturbation. The resulting shifts on the energy splitting between $|5/2, +5/2\rangle_g$ and $|5/2, -5/2\rangle_g$ can be expressed as

$$\Delta E := \delta E_{+5/2}^{(2)} - \delta E_{-5/2}^{(2)} = \frac{5}{2} \cdot \left(\frac{\Omega_B}{\Omega_T}\right)^2 \cdot \frac{\Omega_B\theta^2}{\left(\frac{\Omega_B}{\Omega_T} + 4\right)\left(\frac{\Omega_B}{\Omega_T} - 4\right)}, \quad (2)$$

where $\delta E_{m_F}^{(2)}$ denotes the second-order perturbation correction to the energy level of $|5/2, m_F\rangle_g$. Derivation of equation (2) is presented in the Methods. It can be seen that on opposite sides of the critical ratio $\Omega_B/\Omega_T = 4$, the sign of ΔE reverses, resulting in curvature inversion of the R_L surface. At the critical ratio of $\Omega_B/\Omega_T = 4$, due to the degeneracy between $|5/2, -5/2\rangle_g$ and $|5/2, -3/2\rangle_g$, the above perturbation treatment is no longer valid, and the cat state can no longer be preserved.

To evaluate the robustness of the comagnetometer against environmental magnetic-field fluctuations, we define the suppression factor η as

$$\eta(B) = \left| \frac{\Delta B}{B} / \frac{\Delta R_L}{R_L} \right|. \quad (3)$$

From 50 mG to 100 mG, the magnetic field changes by a factor of two and so do the Larmor frequencies of each isotope, yet R_L remains constant at the 10-ppm level. We derive the suppression factor $\eta > 3 \times 10^4$ (95% confidence level) based on the two R_L values and their corresponding uncertainties.

Residual vector shift. To minimize the vector shifts, the optical lattice is tuned to be as close to linearly polarized as possible. This is done by optimizing the coherence of the Ramsey interferometry of ^{171}Yb . During polarization tuning, the bias magnetic field is temporarily redirected to the horizontal propagation direction of the optical lattice (y axis) and its strength lowered to 1 mG, both measures taken to deliberately enhance the decoherence effect due to the inhomogeneous vector shifts in the lattice. In this way, the residual circular polarization component, as measured by the Stokes parameter S_3 , is tuned to be less than 0.1%. According to the calculated vector polarizabilities of ^{171}Yb and ^{173}Yb , the vector shifts due to the residual circular polarization can be regarded as effective magnetic fields less than 100 μG along the y direction. At a bias field of 50 mG in the z direction, the relative deviation in R_L due to the residual vector shift is estimated to be less than 3×10^{-7} , which is negligible compared with the current measurement uncertainty of 3×10^{-6} . When the field is increased to 100 mG, the deviation drops further down to 8×10^{-8} .

Ratio of nuclear magnetic moments

We have investigated systematic uncertainties in the measurements of R_L . After aligning the 100-mG magnetic field to be parallel to the

polarization direction of the optical lattice, we measure R_L and vary the lattice power by more than a factor of five. No dependence of R_L on light power is observed within the measurement uncertainties (Fig. 3a). A weighted average across different lattice powers yields $R_L = 0.726077(2)$. Separately, we measure R_L and vary the density ratio of ^{171}Yb over ^{173}Yb by a factor of seven. This is to investigate possible collisional shifts that are likely to be density dependent. Again, no dependence of R_L on the density ratio is observed (Fig. 3b). The average value across the different density ratios is $0.726072(4)$. The Earth's rotation frequency of $2\pi \times 11.6 \mu\text{Hz}$ is negligible compared with the measurement uncertainty of around $2\pi \times 100 \mu\text{Hz}$ for the Larmor frequencies.

Since the two independently determined R_L values agree, we take a weighted average of the two, yielding a final value of $R_L = 0.726076(3)$ and the ratio of the nuclear magnetic moments between ^{171}Yb and ^{173}Yb as $R_{\text{NMM}} = \mu_{I,171}/\mu_{I,173} = -0.726076(3)$. As shown in Table 1, our result is consistent with the reported values from prior works^{31–33}, and improving the precision by an order of magnitude.

Discussion

With its strong suppression of magnetic-field noise and tight atom localization within the lattice sites, our cold-atom comagnetometer enables sensitive searches for exotic interactions beyond the Standard Model, particularly at short interaction ranges of 1–100 μm . Among the many new-physics interactions proposed^{3,34}, we present two examples highlighting the capability of this system.

The comagnetometer is used to search for P- and T-violating spin-mass couplings, also known as monopole–dipole interactions^{12,13,19,35}. In this search, with ^{171}Yb and ^{173}Yb atoms as spin sources, the value of R_L is monitored as a dense mass source and brought into proximity with the atoms. By using optical tweezers to position the atoms, the atom–mass separation can be precisely controlled down to a few micrometres. For example, with one pair of ^{171}Yb – ^{173}Yb atoms, a minimum atom–mass separation of 5 μm and 30 days of data accumulation, the sensitivity to spin–mass couplings at the micrometre-scale range can be improved by three orders of magnitude over the current best experimental limits³⁶.

Furthermore, we aim to improve the search sensitivity of EDMs using cold atoms^{5,37}. The current EDM experiments on ^{171}Yb (ref. 5) are limited by magnetic-field noises, particularly the Johnson noise in adjacent electrodes. A ^{171}Yb – ^{173}Yb cold-atom comagnetometer can effectively suppress such noise effects, thereby boosting the EDM search sensitivity.

Online content

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Methods

Zeeman and tensor light shifts of ^{173}Yb

The Hamiltonian in equation (1) is rewritten below, treating the σ_x term as a perturbation:

$$\begin{aligned}\mathcal{H} &= \mathcal{H}_0 + \theta \mathcal{H}' \\ &= \Omega_B \sigma_z + \Omega_T \sigma_z^2 + \theta \Omega_B \sigma_x \quad (\theta \ll 1).\end{aligned}\quad (4)$$

The perturbed eigenenergies are

$$E_{m_f} = E_{m_f}^{(0)} + \theta \langle \psi_{m_f}^{(0)} | \mathcal{H}' | \psi_{m_f}^{(0)} \rangle + \sum_{i \neq m_f} \theta^2 \frac{\langle \psi_i^{(0)} | \mathcal{H}' | \psi_{m_f}^{(0)} \rangle^2}{E_{m_f}^{(0)} - E_i^{(0)}} + O(\theta^3), \quad (5)$$

$$E_{m_f}^{(0)} = m_f \Omega_B + m_f^2 \Omega_T, \quad (6)$$

where the θ^2 term corresponds to $\delta E_{m_f}^{(2)}$ in equation (2). When $\Omega_B/\Omega_T = 4$, the unperturbed energy levels of $|5/2, -5/2\rangle_g$ and $|5/2, -3/2\rangle_g$ are degenerate.

Here the first-order perturbation term is always zero because the σ_x term corresponds to a subdiagonal matrix. For the spin-5/2 system in ^{173}Yb , the σ_x matrix is

$$\sigma_x = \begin{pmatrix} 0 & \frac{\sqrt{5}}{2} & & & & \\ \frac{\sqrt{5}}{2} & 0 & \frac{\sqrt{8}}{2} & & & \\ & \frac{\sqrt{8}}{2} & 0 & \frac{\sqrt{9}}{2} & & \\ & & \frac{\sqrt{9}}{2} & 0 & \frac{\sqrt{8}}{2} & \\ & & & \frac{\sqrt{8}}{2} & 0 & \frac{\sqrt{5}}{2} \\ & & & & \frac{\sqrt{5}}{2} & 0 \end{pmatrix}. \quad (7)$$

To calculate the Larmor frequency of the cat state of ^{173}Yb , we consider the second-order perturbation on the two populated stretched states as $|5/2, \pm 5/2\rangle_g$:

$$\delta E_{+5/2}^{(2)} = \theta^2 \cdot \frac{5}{4} \cdot \frac{\Omega_B^2}{\Omega_B + 4\Omega_T}, \quad (8)$$

$$\delta E_{-5/2}^{(2)} = \theta^2 \cdot \frac{5}{4} \cdot \frac{\Omega_B^2}{-\Omega_B + 4\Omega_T}. \quad (9)$$

Subtracting the two energy shifts yields equation (2).

Calibration of Ω_B/Ω_T

We perform precession measurements on ^{173}Yb to calibrate the Ω_B/Ω_T values. Since the cat state is immune to tensor shifts (Ω_T), we use a coherent spin state instead, whose spin is fully polarized along a transverse direction. As a result, it encompasses all m_f states.

Although the lattice is kept linearly polarized along the z axis, its power is reduced to 8 W to mitigate decoherence effects due to tensor shifts. Meanwhile, a bias magnetic field of 23.5 mG is applied along the x direction. The ^{173}Yb atoms are first pumped into $|5/2, +5/2\rangle_g$ and then allowed to precess under the influence of both Zeeman and tensor shifts, $\mathcal{H}_{\text{calib}} = \Omega_B \sigma_x + \Omega_T \sigma_z^2$. After a variable precession duration (0–2,500 ms), the population signal S_z is measured along the z axis (Extended Data Fig. 1).

We simulate the precession process of ^{173}Yb using the calibration Hamiltonian ($\mathcal{H}_{\text{calib}}$) and compare the results with experimental data. In the simulation, the contribution of each m_f state to the population signal is weighted by both the Clebsch–Gordan coefficient and the

probe-beam detuning for the relevant transition (Extended Data Fig. 2b). Specifically, when intending to probe the population in $|5/2, +5/2\rangle_g$, the detection probabilities for all $|5/2, m_f\rangle_g$ states are 1.00, 0.58, 0.31, 0.16, 0.08 and 0.03 (m_f from +5/2 to –5/2). Furthermore, the spatial distribution of the tensor field across the optical trap is incorporated in the model. By comparing simulated results with the experimental data, we extract $\Omega_B/\Omega_T = 3.5$ under the given magnetic field and lattice intensity condition.

It should be noted that the detection issue described above, involving all m_f states, is specific to the coherent spin state used for the calibration. By contrast, the cat states prepared elsewhere in the experiment involve only $|5/2, \pm 5/2\rangle_g$ states, which have maximally different detection probabilities (Extended Data Fig. 2b) and are, thus, clearly resolvable.

Atom sample detection

A circularly polarized laser probe beam along the z direction runs through the cold-atom ensemble and is detected by a complementary metal–oxide–semiconductor camera. Absorption images are taken on the atoms for population measurements. The state read-out of atoms is performed via the differential detection of two stretched states. For ^{173}Yb atoms, the probe beam resonantly excites the cycling transition $|1/2, +1/2\rangle_g \rightarrow |3/2, +3/2\rangle_e$ (Extended Data Fig. 2a). Meanwhile, the $|1/2, -1/2\rangle_g \rightarrow |3/2, +1/2\rangle_e$ transition is off-resonance by ten times the natural linewidth (10Γ) due to tensor shifts on the excited states. As a result, the cycling transition can be repeated 1,200 times on average before a spin flip occurs. This spin selectivity ensures that the spin state is detected with an efficiency approaching 100%. For ^{173}Yb atoms, the relevant transitions are $|5/2, +5/2\rangle_g \rightarrow |7/2, +7/2\rangle_e$ and $|5/2, -5/2\rangle_g \rightarrow |3/2, -3/2\rangle_e$ (Extended Data Fig. 2b). In addition to a comparable difference of tensor shifts ($\sim 20\Gamma$), the branching ratios of these two transitions differ by a factor of 21, thereby further enhancing the spin selectivity.

Data availability

Source data for Figs. 1c, 2b,d and 3 and Extended Data Fig. 1 are available via Figshare at <https://doi.org/10.6084/m9.figshare.32043486> (ref. 38).

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- Zhang, J.-L. et al. Cold-atom comagnetometry via optical control of spin states. *Figshare* 10.6084/m9.figshare.32043486 (2026).

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Author contributions

J.-L.Z. and W.-T.L. constructed the experimental apparatus and performed the experiments and simulations. Y.A.Y. contributed to the apparatus during the early stages. Y.-Q.W. made contributions to the theory part. J.-L.Z., W.-T.L., Y.-Q.W., T.X. and Z.-T.L. carried out the data analysis and wrote the manuscript.

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Competing interests

The authors declare no competing interests.

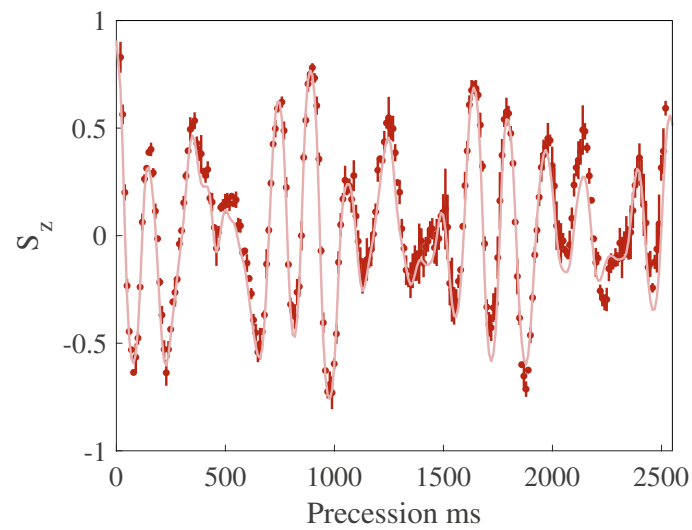
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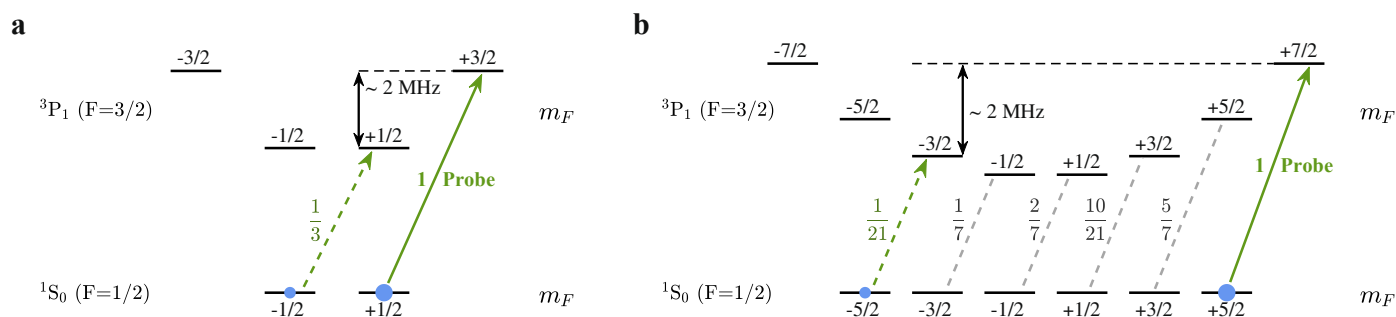
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Extended Data Fig. 1 | Precession of a coherent spin state of ^{173}Yb under $B_x = 23.5$ mG and 8 W lattice power. Error bars: one standard deviation ($1-\sigma$) derived from 3 repeated measurements. The solid curve represents simulation results.



Extended Data Fig. 2 | Transitions used to measure spin-state populations in ^{171}Yb (a) and ^{173}Yb (b). The green solid arrows indicate cycling transitions used for population measurements. The green dashed arrows indicate spin-flip transitions suppressed by large detunings. The gray dashed lines

indicate transitions absent in cat-state measurements due to the lack of atomic population in corresponding m_F -states. Values adjacent to all transition lines denote the squares of the corresponding Clebsch-Gordan coefficients.

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